High Temperature, High Purity Induction Furnace

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Solid Oxide Fuel Cells

Solid oxide fuel cells (SOFC) are a viable source of clean energy that can greatly improve how we generate power. The challenge that these fuel cells face is high operating temperatures of around 1000C [1]. To reduce the operating temperature of SOFC, research is being performed to characterize the ionic conductivity of individual grain boundaries within the electrolyte of the SOFC.

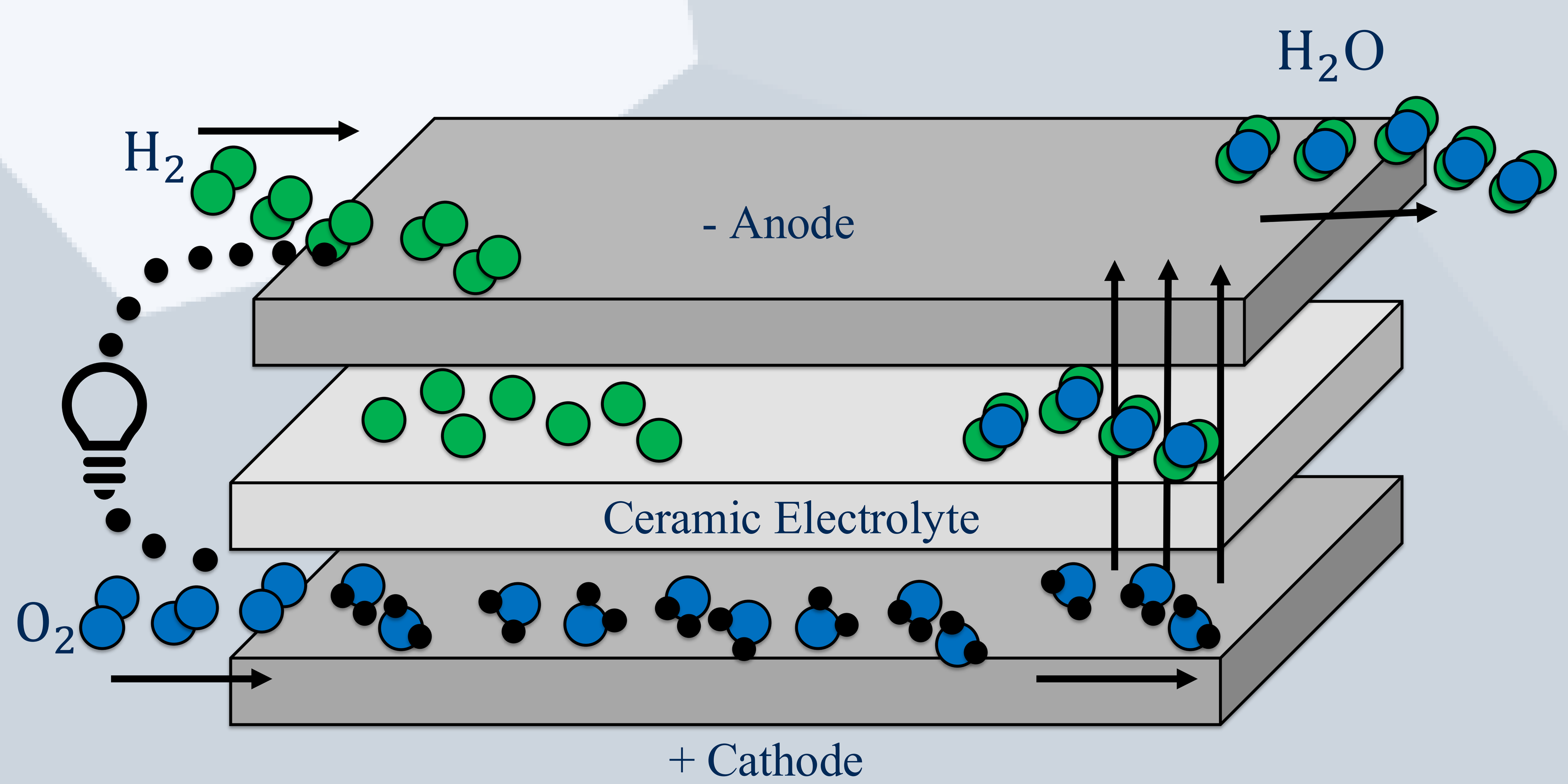

Figure 1: Conversion of chemical to electrical energy in a solid oxide fuel cell system.

In order to characterize the ionic conductivity, the grain boundary impedance is measured using microcontact electrochemical impedance spectroscopy (EIS) which requires small contacts, about 100 microns in size, to be placed on individual grains on the electrolyte. In order to place this contact on an individual grain and make the measurements needed, a grain size of approximately 250 microns is required. However, typical SOFC electrolytes have grain sizes on the order of 1 micron which makes these measurements very difficult [1][2].

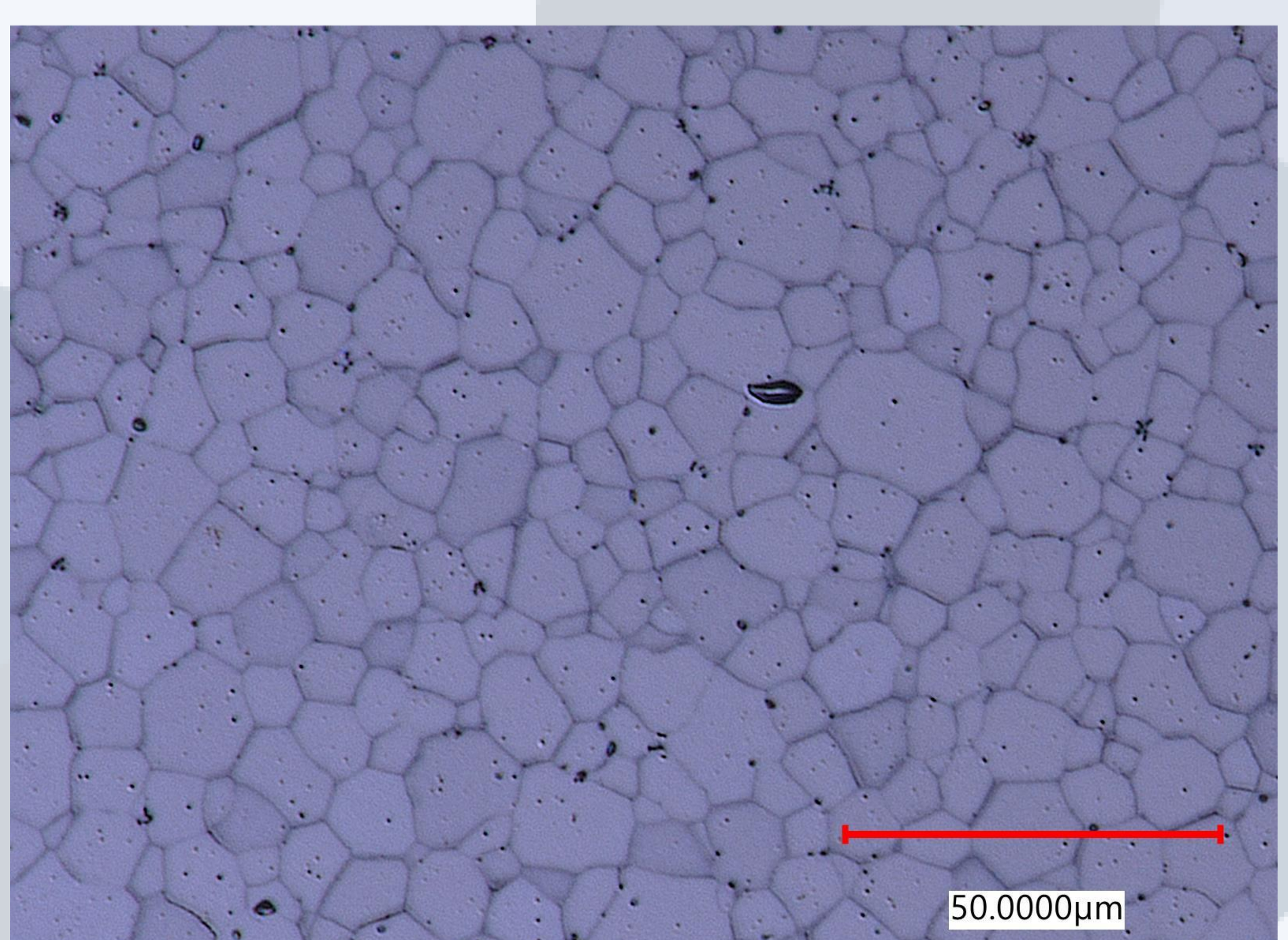

Figure 2: Thermally etched YSZ with an average grain size of ~10 microns.

Induction Furnace Design

While growing grains is possible it is often difficult in refractory ceramics like the SOFC electrolyte. This has led to the creation of a high-temperature induction furnace capable of operating at 2500C for extended periods. This furnace operates under inert gas (Ar) conditions, which ensures the purity of the sample. Due to this high operating temperature significant effort has been put into finding a suitable susceptor, avoiding chemical reactions, limiting, and dealing with heat transfer to support materials, and vibration isolation. The closest commercially available furnace that meets these specifications operates 500C lower than needed and is a significant investment [3].

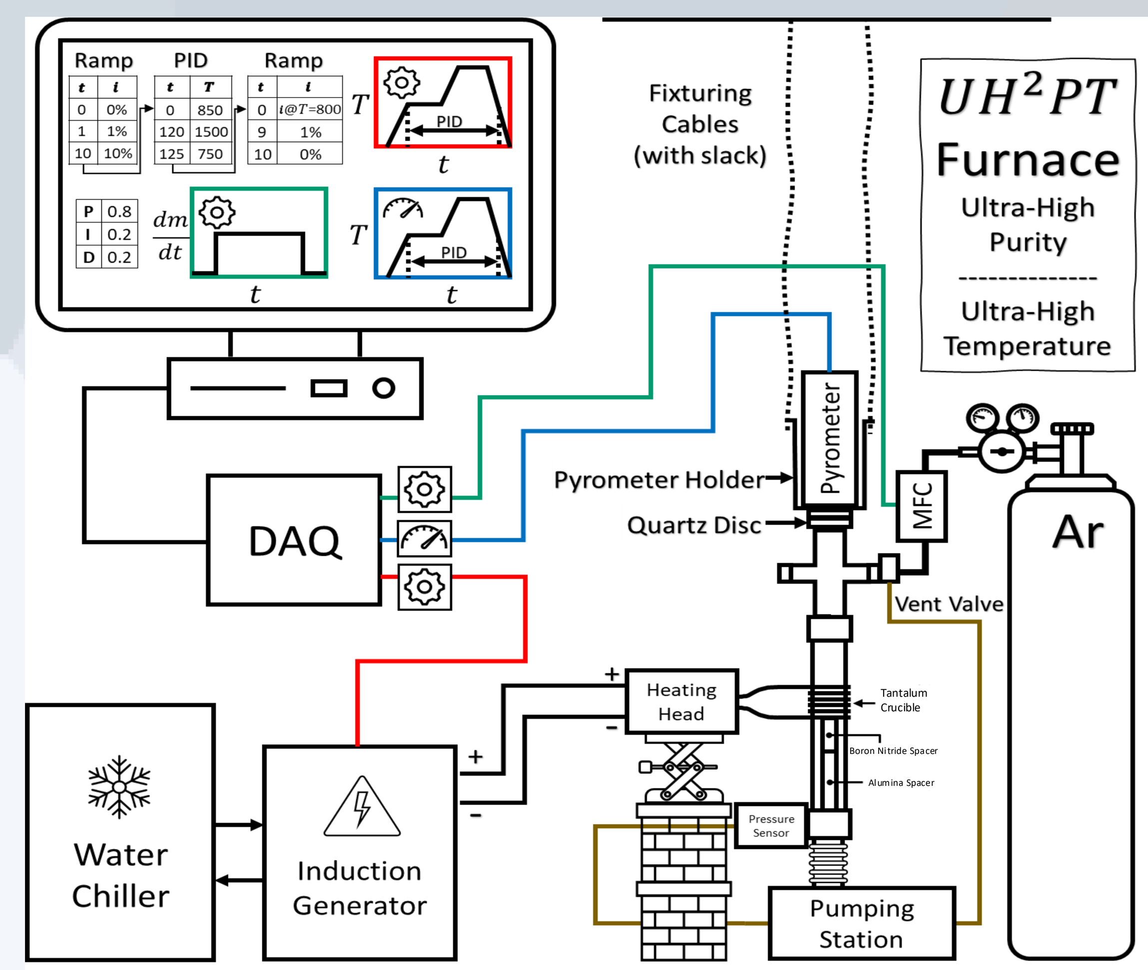

Figure 3: Schematic of the induction furnace setup. Figure credit: Sterling Baird

Among the selected materials, perhaps the most important is the boron nitride as it is not a susceptor to induction and has a high enough melting temperature and low enough thermal conductivity to protect the rest of the system from the extreme temperatures. Unlike many other materials, the boron nitride has not exhibited any significant chemical reactions at 2500C.

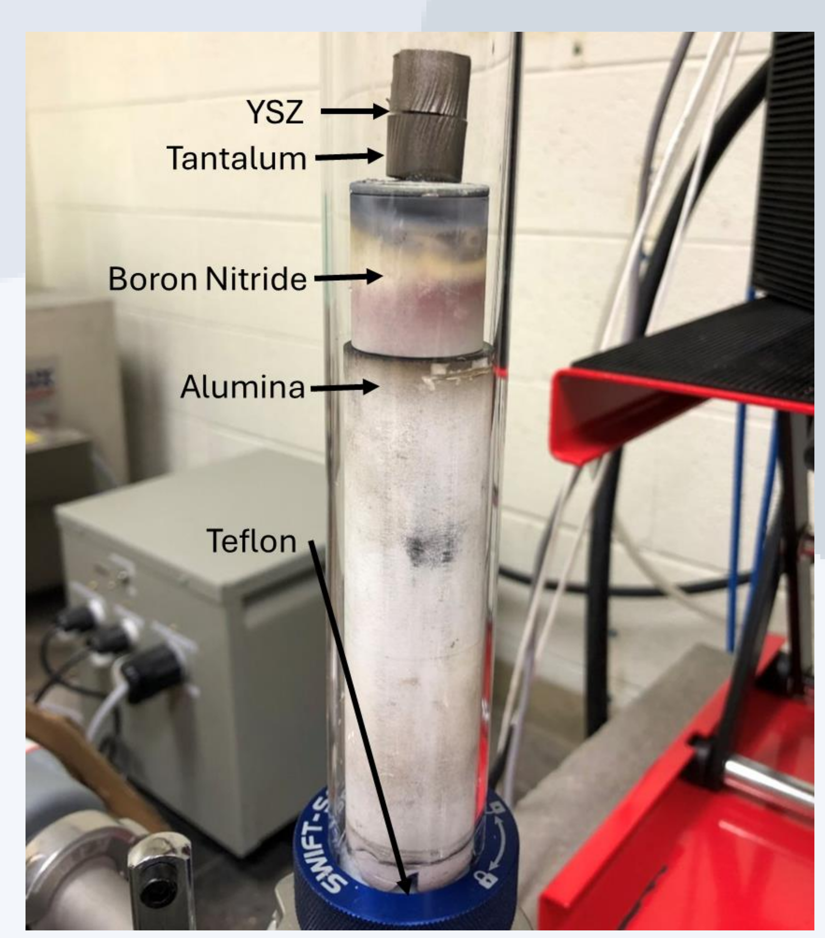

Figure 4: Materials used to grow grains in a sample of YSZ at 2500C

Grain Growth

Using a sample of 8 mol% yttria-stabilized zirconia (YSZ), this furnace has demonstrated the ability to grow grains from an average size of 20 to 90 microns in 45 minutes while operating at 2500C. A commercially available furnace [4] whose max operating temperature is 1600C took over 228 hours to grow grains from 10 to 80 microns. Longer tests of around 3.5 hours have yielded much larger grains as seen in Figure 5. Figure 5 also shows impurities in the YSZ sample due to diffusion bonding that occurs between the tantalum susceptor and the YSZ sample. To prevent this diffusion bonding from occurring a diffusion barrier of YSZ power has been placed between the lower tantalum disk seen in Figure 4 and the YSZ sample. The upper tantalum disk has been removed. Using this setup, longer high temperature tests can be preformed without as many impurities in the sample.

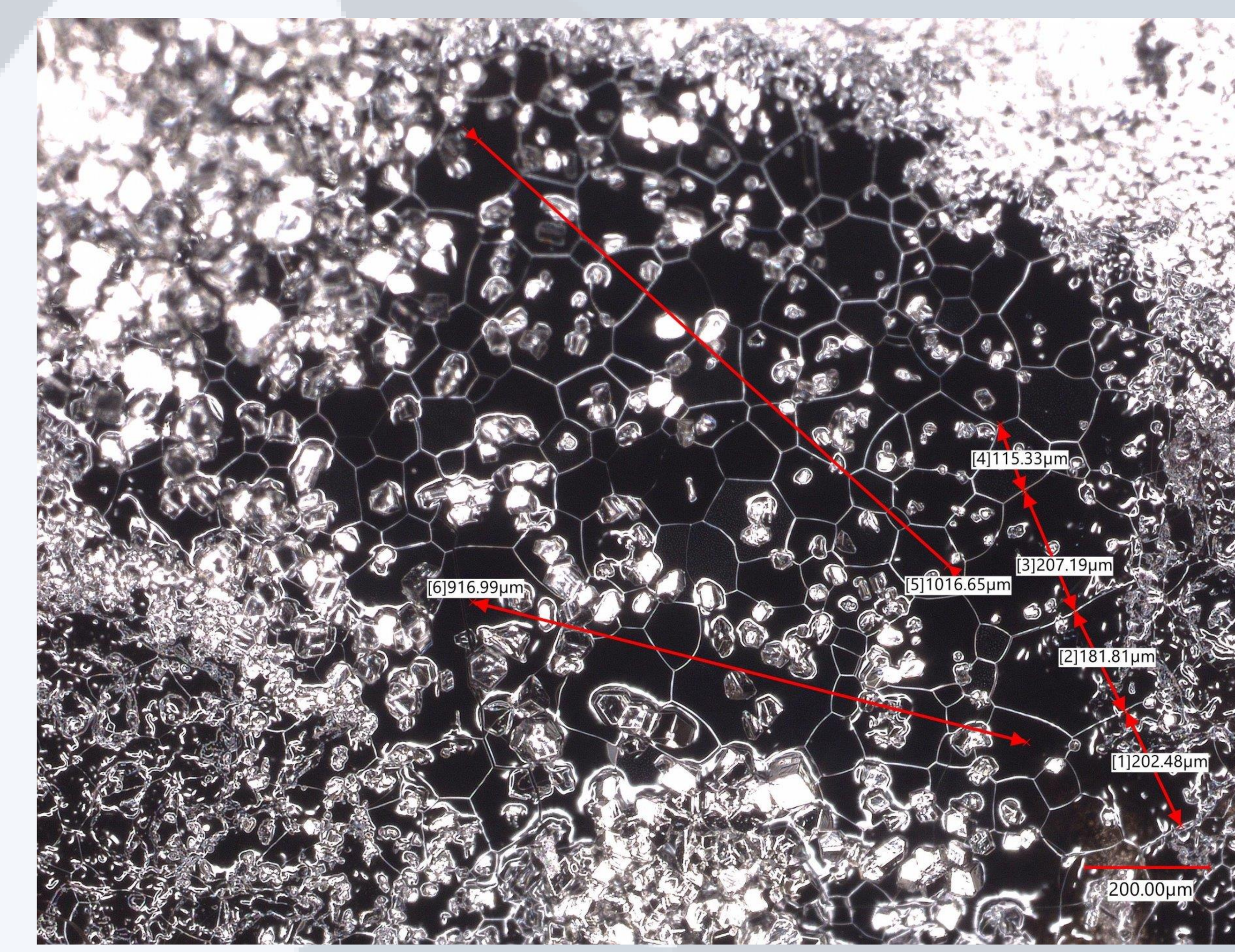

Figure 5. Large grain YSZ after 3.5 hours at 2500C. Impurities in the sample still prove to be an issue.

Conclusion

This high-temperature furnace enables SOFC research by allowing us to produce samples with the grain size required to preform EIS measurements. This furnace achieves these results while being a fraction of the cost of a commercial inert induction furnace, making the study of SOFCs more accessible and viable.

References

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